

Outer-In Decimal Digit Permutations with Equal Prime Support

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Abstract

We study the decimal digit transformation obtained by reading a number from the two ends inward. For

$$n = d_1 d_2 \cdots d_k,$$

define

$$T(n) = d_1 d_k d_2 d_{k-1} d_3 d_{k-2} \cdots .$$

We seek positive integers n for which $n \neq T(n)$ and n and $T(n)$ have the same set of distinct prime divisors. The sequence considered here is restricted to integers not ending in 0. We prove an exact criterion in terms of $g = \gcd(n, T(n))$ and the coprime reduced pair $n/g, T(n)/g$. This gives a factorization-light exact test and leads to an exhaustive computation through 10^{10} . Exactly 22 integers satisfy the condition in the stated search domain. We also prove three infinite repeated-block families generated by $224 \mapsto 242$, $448 \mapsto 484$, and $117 \mapsto 171$.

1 Definition and motivation

Let

$$n = d_1 d_2 \cdots d_k$$

be the usual decimal expansion of a positive integer, with $d_1 \neq 0$. Define the outer-in transformation T by taking the first digit, then the last digit, then the second digit, then the second-to-last digit, and so on:

$$T(n) = d_1 d_k d_2 d_{k-1} d_3 d_{k-2} \cdots .$$

For example,

$$T(12345678) = 18273645, \quad T(12345) = 15243, \quad T(3024) = 3402.$$

For a positive integer n , write

$$\text{supp}(n) = \{p : p \text{ is prime and } p \mid n\}$$

and

$$\text{rad}(n) = \prod_{p \mid n} p.$$

Thus $\text{rad}(n) = \text{rad}(m)$ is equivalent to equality of the sets of distinct prime divisors of n and m .

The sequence studied here consists of the integers satisfying

$$n \neq T(n), \quad \text{rad}(n) = \text{rad}(T(n)), \tag{1}$$

together with the explicit normalization

$$n \not\equiv 0 \pmod{10}. \tag{2}$$

Condition (2) is part of the definition of the sequence; it is not needed in order for $T(n)$ itself to be defined.

The condition $n \neq T(n)$ excludes the fixed points of this particular digit permutation. It is not the same as requiring a number to be non-palindromic.

2 An exact criterion

Theorem 1. *Let*

$$g = \gcd(n, T(n)), \quad n = ag, \quad T(n) = bg,$$

where $\gcd(a, b) = 1$. Then

$$\text{rad}(n) = \text{rad}(T(n))$$

if and only if

$$\text{rad}(ab) \mid g. \tag{3}$$

Equivalently,

$$a \text{ rad}(ab) \mid n. \tag{4}$$

Proof. Assume first that $\text{rad}(n) = \text{rad}(T(n))$. If a prime p divides a , then $p \mid n$ and hence $p \mid T(n) = bg$. Since $\gcd(a, b) = 1$, $p \nmid b$, so $p \mid g$. Interchanging a and b shows that every prime divisor of b also divides g . Hence every prime dividing ab divides g , proving (3).

Conversely, suppose $\text{rad}(ab) \mid g$. Every prime divisor of a or b then already divides g . Consequently the prime divisors of $ag = n$ are exactly the prime divisors of g , and the same is true of $bg = T(n)$. Therefore

$$\text{rad}(n) = \text{rad}(T(n)).$$

The equivalent form (4) follows from $n = ag$. □

This criterion is the basis of the exhaustive search. It avoids independently factoring the two large integers n and $T(n)$: after computing one gcd, it is enough to test whether every prime divisor of the two reduced factors already occurs in g .

3 Exhaustive computation

The search constructs $T(n)$ directly from the decimal digits. Before the exact test, necessary support checks are performed for

$$2, 5, 7, 11, 13, 17, 19, 23.$$

A candidate is rejected if one of these primes divides exactly one of n and $T(n)$. The primes 3 and 9 require no analogous test because a digit permutation preserves the digit sum.

Every surviving candidate is tested using Theorem 1:

$$g = \gcd(n, T(n)), \quad a = n/g, \quad b = T(n)/g,$$

followed by the exact divisibility condition $\text{rad}(ab) \mid g$.

The reported computation covers

$$0 < n < 10^{10}, \quad n \not\equiv 0 \pmod{10}.$$

The historical computation had two phases: an exhaustive search below 10^9 , followed by a complete search of the ten-digit interval

$$10^9 \leq n < 10^{10}.$$

The ten-digit phase used an audited 64-bit implementation. Since

$$10^{10} < 2^{64},$$

all values in that interval are represented exactly by the implementation.

The resulting 22-term initial segment is

3024, 6048, 10143, 20286, 27864, 117216, 189216, 224224,
 333234, 352872, 387585, 448448, 666468, 3892032, 3928824,
 137539323, 247153125, 321881616, 869148324, 1938285196,
 2240563518, 2952612351.

Proposition 1. *Among positive integers $n < 10^{10}$ with $n \not\equiv 0 \pmod{10}$, exactly 22 integers satisfy (1).*

The first 19 terms occur below 10^9 ; the complete ten-digit interval contains exactly three terms.

Table 1: The 22 verified terms, their images, gcds, and reduced ratios.

n	$T(n)$	$\gcd(n, T(n))$	$n/g : T(n)/g$
3024	3402	378	4:9
6048	6804	756	8:9
10143	13041	1449	7:9
20286	26082	2898	7:9
27864	24768	3096	9:8
117216	161172	14652	8:11
189216	168192	21024	9:8
224224	242242	2002	112:121
333234	343332	10098	33:34
352872	325728	27144	13:12
387585	358875	14355	27:25
448448	484484	4004	112:121
666468	686664	20196	33:34
3892032	3283902	121626	32:27
3928824	3492288	436536	9:8
137539323	133273593	60939	2257:2187
247153125	254271135	395445	625:643
321881616	362116818	40235202	8:9
869148324	846293184	653004	1331:1296
1938285196	1699318528	26551852	73:64
2240563518	2821450356	82983834	27:34
2952612351	2195532261	75708009	39:29

For the three ten-digit pairs, independent factorizations are

$$1938285196 = 2^2 \cdot 73^2 \cdot 90931, \quad 1699318528 = 2^8 \cdot 73 \cdot 90931,$$

$$2240563518 = 2 \cdot 3^5 \cdot 17 \cdot 139 \cdot 1951,$$

$$2821450356 = 2^2 \cdot 3^2 \cdot 17^2 \cdot 139 \cdot 1951,$$

and

$$2952612351 = 3^3 \cdot 13^2 \cdot 29 \cdot 53 \cdot 421,$$

$$2195532261 = 3^2 \cdot 13 \cdot 29^2 \cdot 53 \cdot 421.$$

Thus the equality of prime support in all three ten-digit cases can also be checked directly.

As a small example,

$$3024 = 2^4 \cdot 3^3 \cdot 7, \quad 3402 = 2 \cdot 3^5 \cdot 7,$$

so both numbers have radical 42.

The computational completeness statement is only for the stated range and the nonzero-last-digit restriction.

4 Repeated blocks and infinite families

Let B and C be decimal blocks of the same length m .

Lemma 1. *If*

$$T(BB) = CC, \tag{5}$$

then for every $r \geq 1$,

$$T(B^{2r}) = C^{2r}, \tag{6}$$

where B^{2r} denotes the decimal concatenation of $2r$ copies of B .

Proof. Write

$$B = b_1 b_2 \cdots b_m.$$

In the word B^{2r} , consider the output in consecutive pairs. The first pair is b_1, b_m , the second is b_2, b_{m-1} , and so on through the m -digit block pattern

$$b_1, b_m, b_2, b_{m-1}, \dots$$

Because the block B is repeated, exactly the same $2m$ -digit pattern is produced for each of the r block-pairs. That pattern is $T(BB) = CC$. Hence the complete output is r copies of CC , namely C^{2r} . $\square$

Let $A = [B]$ and $C_0 = [C]$ be the numerical values of the blocks, and define

$$Q_r = 1 + 10^m + 10^{2m} + \cdots + 10^{(2r-1)m}.$$

Then

$$[B^{2r}] = AQ_r, \quad [C^{2r}] = C_0 Q_r. \tag{7}$$

Moreover,

$$10^m + 1 \mid Q_r, \tag{8}$$

because $10^m \equiv -1 \pmod{10^m + 1}$ and Q_r contains an even number of powers.

Let

$$g = \gcd(A, C_0), \quad A = ag, \quad C_0 = bg, \quad \gcd(a, b) = 1.$$

Combining (7), (8), and Theorem 1 gives the following sufficient condition.

Proposition 2. *If*

$$\text{rad}(ab) \mid g(10^m + 1), \tag{9}$$

then every even repetition B^{2r} has the same prime support as the corresponding repetition C^{2r} .

Proof. Since $10^m + 1 \mid Q_r$, condition (9) implies

$$\text{rad}(ab) \mid gQ_r.$$

Applying Theorem 1 to AQ_r and $C_0 Q_r$ proves equality of prime support for every $r \geq 1$. $\square$

4.1 The family $224 \mapsto 242$

We have

$$224 = 2^5 \cdot 7, \quad 242 = 2 \cdot 11^2,$$

so

$$g = 2, \quad a = 112, \quad b = 121, \quad \text{rad}(ab) = 2 \cdot 7 \cdot 11.$$

For $m = 3$,

$$10^3 + 1 = 1001 = 7 \cdot 11 \cdot 13.$$

Hence (9) holds. Therefore, for every $r \geq 1$, the decimal integer obtained by repeating 224 exactly $2r$ times is a solution, and its image under T is the corresponding repetition of 242.

The first member is

$$224224 \mapsto 242242.$$

4.2 The family $448 \mapsto 484$

Similarly,

$$448 = 2^6 \cdot 7, \quad 484 = 2^2 \cdot 11^2,$$

so

$$g = 4, \quad a = 112, \quad b = 121.$$

Again $m = 3$, and $10^3 + 1$ contains both required support primes 7 and 11. Thus every even repetition of 448 is a solution and is mapped to the corresponding even repetition of 484.

4.3 The family $117 \mapsto 171$

Take the 9-digit blocks

$$B = 117117117, \quad C = 171171171.$$

Directly,

$$T(BB) = CC,$$

or

$$T(117117117117117117) = 171171171171171171.$$

Also,

$$\gcd(B, C) = 9009009, \quad B/\gcd(B, C) = 13, \quad C/\gcd(B, C) = 19.$$

Thus the reduced pair is $13 : 19$. Finally,

$$10^9 + 1 = 7 \cdot 11 \cdot 13 \cdot 19 \cdot 52579.$$

Hence (9) holds. Every even repetition of B , equivalently every repetition of 117 exactly $6r$ times, is therefore a solution and is mapped to the corresponding repetition of 171.

The first member of this family is

$$117117117117117117 \mapsto 171171171171171171.$$

5 Relation to previous work

The closest OEIS analogue is A110751, which records integers whose decimal digit reversal has the same prime divisors; its non-palindromic subsequence is A110819 [1, 2]. The present transformation is different: it alternates digits from opposite ends rather than reversing the whole digit string.

A244514 gives another nearby example: it studies integers for which all cyclic permutations of the decimal digits have the same prime divisors [3]. The present problem fixes one specific digit permutation rather than requiring the property for a whole cyclic orbit. A371307 records the analogous smallest nonpalindromic examples for digit reversal in varying bases.

More generally, digit-permutation arithmetic is studied in the theory of permutiples. Holt’s papers treat integers that are nontrivial multiples of permutations of their digits [4, 5]. No multiplicative relation such as $T(n) = qn$ is imposed here; the invariant is equality of the sets of distinct prime divisors.

Searches of the OEIS and web-indexed sources performed for this work through 22 September 2026 did not locate an exact match for the displayed initial terms together with the outer-in/equal-prime-support definition. This is a report of the searches performed, not a formal priority claim.

6 Reproducibility and limitations

The accompanying archive separates the exploratory research history from a clean reproducibility release. The release contains the exhaustive-search source, an independent verifier, the consolidated dataset, the raw ten-digit outputs, the OEIS data, and verification records.

The search program operates on one decimal digit length at a time, so the width used to construct $T(n)$ is fixed during each run. The ten-digit interval is therefore searched as

$$[10^9, 10^{10}).$$

The implementation uses 64-bit unsigned integers, which are sufficient throughout this bounded domain.

The independent verifier reconstructs $T(n)$, computes the gcd and reduced pair, checks the exact criterion of Theorem 1, and checks the expected image for each candidate. It also performs explicit factorization checks for the 22 listed pairs. Those factorizations are used as independent certificates for the present dataset, not as a general-purpose large-integer factoring method.

The exhaustive result should therefore be read exactly as stated:

$$0 < n < 10^{10}, \quad n \not\equiv 0 \pmod{10}.$$

The repeated-block constructions show that the full sequence is infinite, so the 22 terms form a verified finite initial segment rather than a complete enumeration of all terms.

7 Conclusion

The outer-in decimal digit transformation gives a distinct prime-support preservation problem. The reduction

$$\text{rad}(n) = \text{rad}(T(n)) \iff \text{rad}\left(\frac{n}{g} \frac{T(n)}{g}\right) \mid g, \quad g = \gcd(n, T(n)),$$

provides an exact characterization and an efficient test for exhaustive computation.

Under the explicit convention that the input does not end in zero, the exhaustive computation finds exactly 22 solutions below 10^{10} . The repeated-block argument then gives infinite families beginning with

$$224 \mapsto 242, \quad 448 \mapsto 484, \quad 117 \mapsto 171.$$

References

- [1] OEIS Foundation, *A110751: Numbers n such that n and its digital reversal have the same prime divisors*. <https://oeis.org/A110751>.
- [2] OEIS Foundation, *A110819: Non-palindromes in A110751*. <https://oeis.org/A110819>.
- [3] OEIS Foundation, *A244514: Numbers n such that the integers formed by all cyclic permutations of the decimal digits of n have the same prime divisors*. <https://oeis.org/A244514>.
- [4] Benjamin V. Holt, “On Permutiples Having a Fixed Set of Digits,” *INTEGERS* 17 (2017), A20. <https://arxiv.org/abs/1511.02033>.
- [5] Benjamin V. Holt, “A Method for Finding All Permutiples with a Fixed Set of Digits from a Single Known Example,” *INTEGERS* 24 (2024), A88. <https://arxiv.org/abs/2312.02056>.